

• Exercice 21

a) $(A | I_3) = \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & -1 & 3 & 0 & 1 & 0 \\ 4 & 1 & 8 & 0 & 0 & 1 \end{array} \right)$

$\sim_L$ $\begin{array}{l} L_2 \leftarrow L_2 - 2L_1 \\ L_3 \leftarrow L_3 - 4L_1 \end{array} \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & -4 & 0 & 1 \end{array} \right)$

$\sim_L$ $L_3 \leftarrow L_3 + L_2 \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 0 & -1 & -6 & 1 & 1 \end{array} \right)$

On a $\text{rg}(A) = 3$ donc A est inversible.

$(A | I_3) \sim_L \begin{array}{l} L_2 \leftarrow -L_2 \\ L_3 \leftarrow -L_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 6 & -1 & -1 \end{array} \right)$

$\sim_L \begin{array}{l} L_1 \leftarrow L_1 - 2L_2 \\ L_2 \leftarrow L_2 - L_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & -2 & 0 & -3 & 2 & 0 \\ 0 & 1 & 0 & -4 & 0 & 1 \\ 0 & 0 & 1 & 6 & -1 & -1 \end{array} \right)$

$\sim_L L_1 \leftarrow L_1 + 2L_2 \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -11 & 2 & 2 \\ 0 & 1 & 0 & -4 & 0 & 1 \\ 0 & 0 & 1 & 6 & -1 & -1 \end{array} \right)$

Donc A est inversible d'inverse

$$A^{-1} = \begin{pmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{pmatrix}$$

$$b) (B | I_3) = \left(\begin{array}{ccc|ccc} 1 & -2 & 2 & 1 & 0 & 0 \\ 2 & -3 & 6 & 0 & 1 & 0 \\ 1 & 1 & 7 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} \sim \\ L \end{array} \begin{array}{l} L_2 \leftarrow L_2 - 2L_1 \\ L_3 \leftarrow L_3 - L_1 \end{array} \left(\begin{array}{ccc|ccc} 1 & -2 & 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & 3 & 5 & -1 & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} \sim \\ L \end{array} \begin{array}{l} L_3 \leftarrow L_3 - 3L_2 \end{array} \left(\begin{array}{ccc|ccc} 1 & -2 & 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & 0 & -1 & 5 & -3 & 1 \end{array} \right)$$

On a $\text{rg}(B) = 3$ donc B est inversible.

$$(B | I_3) \sim \begin{array}{l} L \\ L_3 \leftarrow -L_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & -2 & 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & 0 & 1 & -5 & 3 & -1 \end{array} \right)$$

$$\begin{array}{l} \sim \\ L \end{array} \begin{array}{l} L_1 \leftarrow L_1 - 2L_3 \\ L_2 \leftarrow L_2 - 2L_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & -2 & 0 & 11 & -6 & -2 \\ 0 & 1 & 0 & 8 & -5 & -2 \\ 0 & 0 & 1 & -5 & 3 & 1 \end{array} \right)$$

$$\begin{array}{l} \sim \\ L \end{array} \begin{array}{l} L_1 \leftarrow L_1 + 2L_2 \end{array} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 27 & -16 & -6 \\ 0 & 1 & 0 & 8 & -5 & -2 \\ 0 & 0 & 1 & -5 & 3 & 1 \end{array} \right)$$

Donc B est inversible d'inverse

$$B^{-1} = \begin{pmatrix} 27 & -16 & -6 \\ 8 & -5 & -2 \\ -5 & 3 & 1 \end{pmatrix}$$

$$\underline{c)} \quad (C|I_3) = \left(\begin{array}{ccc|ccc} 1 & 3 & -4 & 1 & 0 & 0 \\ 1 & 5 & -1 & 0 & 1 & 0 \\ 3 & 13 & -6 & 0 & 0 & 1 \end{array} \right)$$

$$\sim_L \begin{array}{l} L_2 \leftarrow L_2 - L_1 \\ L_3 \leftarrow L_3 - 3L_1 \end{array} \left(\begin{array}{ccc|ccc} 1 & 3 & -4 & 1 & 0 & 0 \\ 0 & 2 & 3 & -1 & 1 & 0 \\ 0 & 4 & 6 & -3 & 0 & 1 \end{array} \right)$$

$$\sim_L \begin{array}{l} L_3 \leftarrow L_3 - 2L_2 \end{array} \left(\begin{array}{ccc|ccc} 1 & 3 & -4 & 1 & 0 & 0 \\ 0 & 2 & 3 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & -2 & 1 \end{array} \right)$$

On a $\text{rg}(C) = 2 \neq 3$ donc la matrice C n'est pas inversible.

$$\underline{d)} \quad (D|I_3) = \left(\begin{array}{ccc|ccc} i & -1 & 2i & 1 & 0 & 0 \\ 2 & 0 & 2 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\sim_L \begin{array}{l} L_1 \leftarrow \frac{1}{i} L_1 \end{array} \left(\begin{array}{ccc|ccc} 1 & i & 2 & -i & 0 & 0 \\ 2 & 0 & 2 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\sim_L \begin{array}{l} L_2 \leftarrow L_2 - 2L_1 \\ L_3 \leftarrow L_3 + L_1 \end{array} \left(\begin{array}{ccc|ccc} 1 & i & 2 & -i & 0 & 0 \\ 0 & -2i & -2 & 2i & 1 & 0 \\ 0 & i & 3 & -i & 0 & 1 \end{array} \right)$$

$$\sim_L \begin{array}{l} L_2 \leftarrow \frac{1}{-2i} L_2 \\ L_3 \leftarrow \frac{1}{i} L_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & i & 2 & -i & 0 & 0 \\ 0 & -2 & 2i & 2 & -i & 0 \\ 0 & 1 & -3i & -1 & 0 & -i \end{array} \right)$$

$$(D|I_3) \underset{L}{\sim} \begin{array}{l} L_2 \leftrightarrow L_3 \\ L_3 \leftrightarrow L_2 \end{array} \left(\begin{array}{ccc|ccc} 1 & i & 2 & -i & 0 & 0 \\ 0 & 1 & -3i & -1 & 0 & -i \\ 0 & -2 & 2i & 2 & -i & 0 \end{array} \right)$$

$$\underset{L}{\sim} \begin{array}{l} L_3 \leftarrow L_3 + 2L_2 \end{array} \left(\begin{array}{ccc|ccc} 1 & i & 2 & -i & 0 & 0 \\ 0 & 1 & -3i & -1 & 0 & -i \\ 0 & 0 & -4i & 0 & -i & -2i \end{array} \right)$$

On a $\text{rg}(D) = 3$ donc D est inversible.

$$(D|I_3) \underset{L}{\sim} \begin{array}{l} L_3 \leftarrow \frac{1}{-4i} L_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & i & 2 & -i & 0 & 0 \\ 0 & 1 & -3i & -1 & 0 & -i \\ 0 & 0 & 1 & 0 & \frac{1}{4} & \frac{1}{2} \end{array} \right)$$

$$\underset{L}{\sim} \begin{array}{l} L_1 \leftarrow L_1 - 2L_3 \\ L_2 \leftarrow L_2 + 3iL_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & i & 0 & -i & -\frac{1}{2} & -1 \\ 0 & 1 & 0 & -1 & \frac{3}{4}i & \frac{1}{2}i \\ 0 & 0 & 1 & 0 & \frac{1}{4} & \frac{1}{2} \end{array} \right)$$

$$\underset{L}{\sim} \begin{array}{l} L_1 \leftarrow L_1 - iL_2 \end{array} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & \frac{1}{4} & -\frac{1}{2} \\ 0 & 1 & 0 & -1 & \frac{3}{4}i & \frac{1}{2}i \\ 0 & 0 & 1 & 0 & \frac{1}{4} & \frac{1}{2} \end{array} \right)$$

Donc D est inversible d'inverse

$$D^{-1} = \begin{pmatrix} 0 & \frac{1}{4} & -\frac{1}{2} \\ -1 & \frac{3}{4}i & \frac{1}{2}i \\ 0 & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$$